MAE 3240 Homework

Name: Zachary Snarr

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Problem Set Number: #3

Due Date: 2/24/26

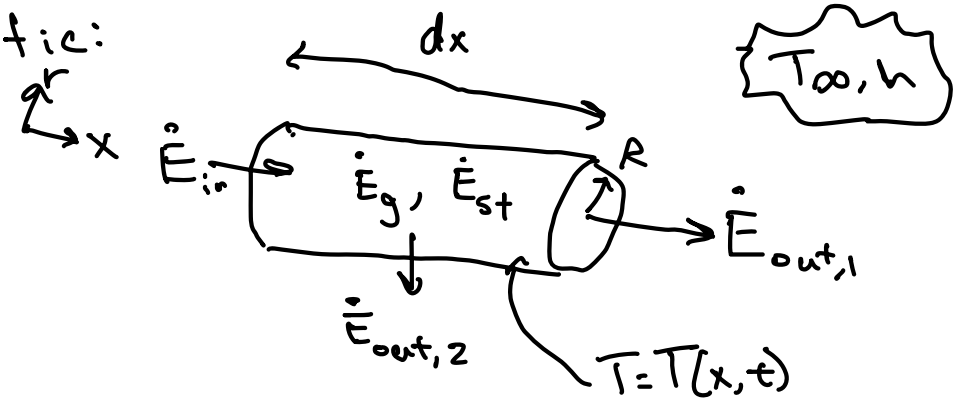
Acknowledgement:

No help

1) Known: radius, uniform volumetric $\dot{q}$, T_∞, h, $T(x,t)$.

Find: a) energy balance eq. as fx of T b) simplify governing diff eq c) boundary & IC's

Schematic:



Properties: h, ρ, c_p

Assumptions: neglect thermal radiation, k ρ c_p are constant

Analysis:

a) $$\dot{E}_{in} + \dot{E}_g = \dot{E}_{out,2} + \dot{E}_{out,1} + \dot{E}_{st}$$

$$-kA\frac{\partial T}{\partial x}\Big|_x + \dot{q}A\,dx = h(2\pi R\,dx)(T - T_\infty) + -kA\frac{\partial T}{\partial x}\Big|_{x+dx} + \rho(A\,dx)c_p\frac{\partial T}{\partial x}$$

b) $$\frac{\partial}{\partial x}\left(k_x\frac{\partial T}{\partial x}\right) + \cancel{\frac{\partial}{\partial y}\left(k_y\frac{\partial T}{\partial y}\right)} + \cancel{\frac{\partial}{\partial z}\left(k_z\frac{\partial T}{\partial z}\right)} + \dot{q} = \rho c_p\frac{\partial T}{\partial t} + \frac{2h}{R}(T - T_\infty)$$

$$\boxed{k\frac{\partial^2 T}{\partial x^2} + \dot{q} = \rho c_p\frac{\partial T}{\partial t} + \frac{2h}{R}(T - T_\infty)}$$

c) BC's

$T(0,t) = T_1$

$-k\frac{\partial T}{\partial x} = h(T(L,t) - T_\infty)$

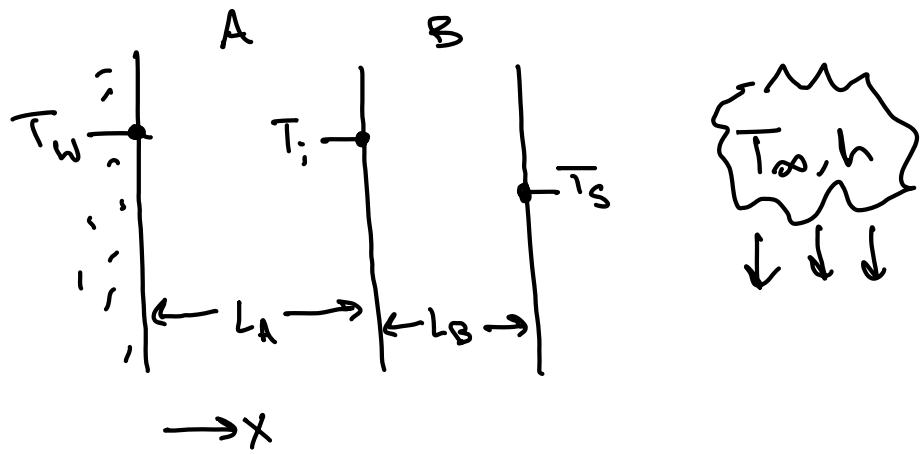
IC's

$T(x,0) = T_i$

2) Known: L_A, uniform volumetric $\dot{q}$, L_B, K_B, T_∞, T_w, T_i

Find: a) T_s of wall b) $\dot{q}$ c) K_A d) plot temp profile

Schematic:



Properties: $K_B = 30 \frac{W}{mK}$, $h_B = 500 \frac{W}{m^2 K}$, K_A

Assumptions: uniform volumetric heat generation $\dot{q}$, steady state conditions

Analysis:

a) $q''_{conv} + q''_{cond} = 0$

$$(T_\infty - T_s) h_B - K_B \left(\frac{T_s - T_i}{L_B} \right) = 0$$

$$T_s h_B + \frac{K_B}{L_B} T_s = \frac{K_B}{L_B} T_i + T_\infty h_B$$

$$T_s = \frac{\frac{K_B}{L_B} T_i + T_\infty h_B}{h_B + \frac{K_B}{L_B}} = \frac{\frac{30}{0.1}(100) + 20(500)}{500 + \frac{30}{0.1}} = 50$$

$$\boxed{T_s = 50°C}$$

b) Wall A control volume

$$q'' = q''_{cond}$$

$$q'' = -k_B\left(\frac{T_s - T_i}{L_B}\right) = -30\left(\frac{50-100}{0.1}\right) = 15000\ W/m^2$$

$$\dot{q} = \frac{15000}{0.3} = 50000\ W/m^3 \qquad \boxed{\dot{q} = 5\times10^4\ W/m^3}$$

c) Simplify heat equation: $\dot{q} = -k_A \frac{d^2T}{dx^2}$

$T(0) = T_w$

$T'(0) = 0$

$T(L_A) = T_i$

$$T'' = -\frac{\dot{q}}{k_A}$$

$$T' = -\frac{\dot{q}}{k_A}x + C_1$$

$$T = -\frac{\dot{q}}{2k_A}x^2 + C_1 x + C_2$$

Apply BC's

$T'(0) = C_1 = 0$

$T(0) = C_2 = T_w$

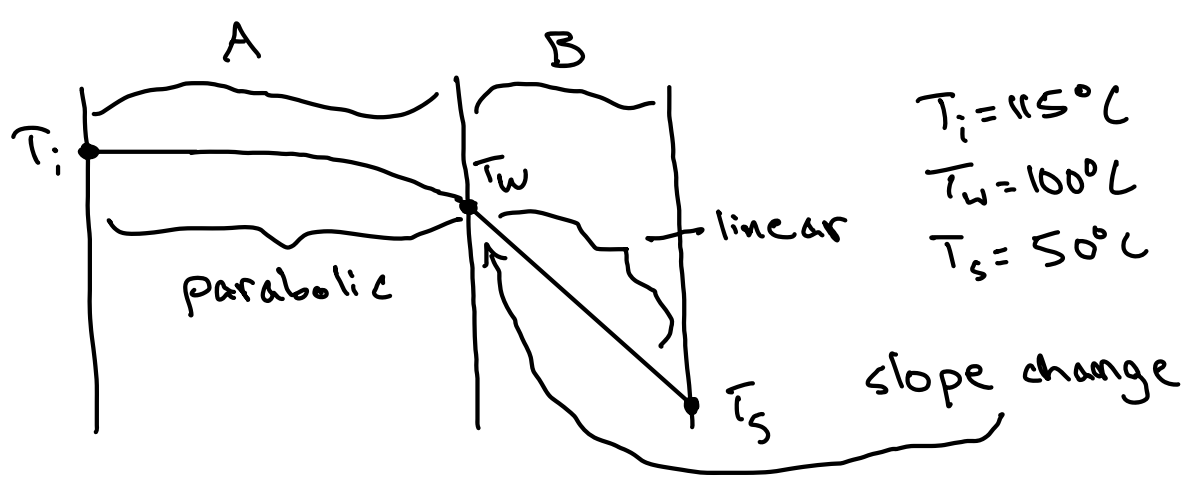
$$\Rightarrow T = -\frac{q}{2k_A}x^2 + T_w$$

$$T(L_A) = -\frac{\dot{q}}{2k_A}L_A^2 + T_w = T_i$$

$$k_A = \frac{\dot{q}L_A^2}{2(T_w - T_i)} = \frac{5\times10^4(0.3^2)}{2(115-100)} = 150\ W/mK$$

$$\boxed{k_A = 150\ W/mK}$$

d)



$T_i = 115°C$

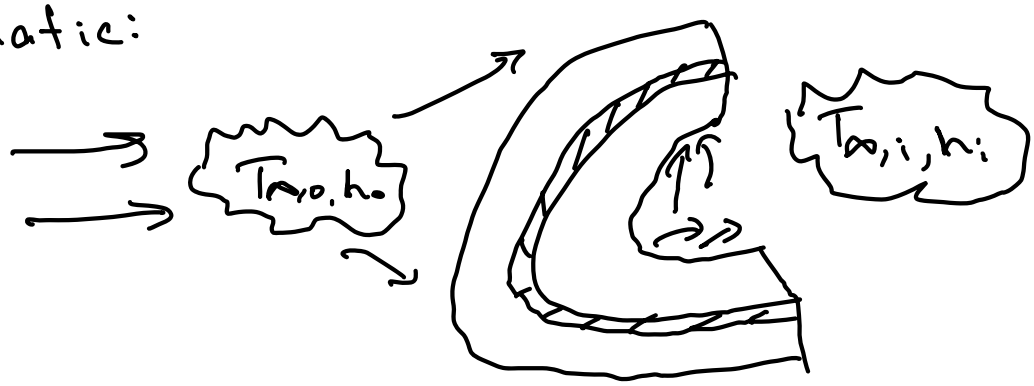
$T_w = 100°C$

$T_s = 50°C$

3) Known: k_b, k_{TBC}, R'', $T_{\infty,o}$, $T_{\infty,i}$, h_o, L_b, T_{max}

Find: a) T_s & is $T_s < T_{max}$ b) same w/ different conditions

Schematic:

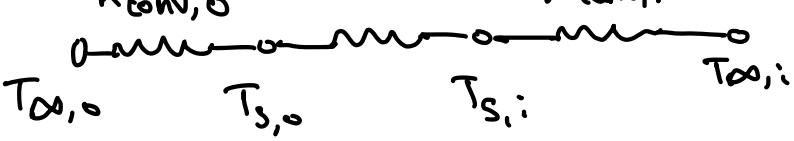


Properties: $k_b = 25 \frac{W}{mK}$, $k_{TBC} = 1 \frac{W}{mK}$, $R'' = 10^{-4} \frac{m^2 K}{W}$

Assumptions: neglect radiation, steady state condition, plane wall approximation, 1D heat cond

Analysis:

a)



$R''_{conv,o} = \frac{1}{h_o}$

$R''_b = \frac{L_b}{k_b}$

$R''_{conv,i} = \frac{1}{h_i}$

$$R''_{tot} = \sum R = \frac{1}{1000} + \frac{0.005}{25} + \frac{1}{500} = 0.0032$$

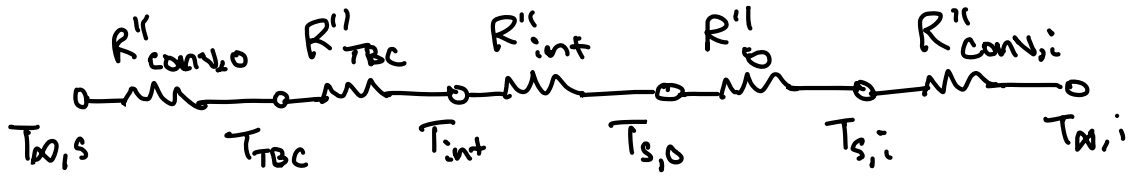
$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{tot}} = \frac{1616 - 400}{0.0032} = 3.8 \times 10^5 \, W$$

$$q'' = (T_{\infty,o} - T_{s,o}) h_o = (1616 - T_{s,o}) 1000 = 3.8 \times 10^5$$

$$\boxed{T_{s,o} = 1236 \, K}$$

$\boxed{1236 > 1200 \text{ so no, the blade can't be kept below } T_{max}}$

b)



$$R''_{tot} = \frac{1}{1000} + \frac{0.0005}{1} + 0.00017 + \frac{0.005}{25} + \frac{1}{500} = 0.0038$$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{tot}} = \frac{1616 - 400}{0.0038} = 3.2\times10^5$$

$$T_{TBC} = T_{\infty,o} - q''R''_{conv,o} = 1616 - 3.2\times10^5(0.001) = 1296$$

$$T_{int} = 1296 - 3.2\times10^5(0.0005) = 1136$$

$$T_{s,o} = 1136 - 3.2\times10^5(0.0001) = 1104\ K$$

$$\boxed{T_{blade,o} = 1104\ K}$$

$1104 < 1200$ ✓ blade can be maintained below T_{max}